

3

NUMBER AND ALGEBRA

ALGEBRA

Algebra is the study of mathematical patterns and rules. The subject can be traced back over 4000 years ago to the ancient Babylonians. Today, all modern technology relies on algebra – the internet, satellites, computers, TVs and all other digital devices rely on the application of algebra.



Chapter outline

	Working mathematically				
3.01 Variables	U	F		R	C
3.02 Creating algebraic expressions	U	F		R	C
3.03 Substitution	U	F	PS		
3.04 Adding and subtracting terms	U	F		R	C
3.05 Multiplying and dividing terms	U	F		R	C
3.06 The index laws*	U	F			
3.07 Expanding expressions	U	F		R	C
3.08 Factorising algebraic terms	U	F		R	
3.09 Factorising expressions	U	F		R	C
3.10 Factorising with negative terms	U	F		R	C
3.11 Expanding binomial products*	U	F		R	C
3.12 Perfect squares*	U	F		R	
3.13 Difference of 2 squares*	U	F		R	C

*YEAR 9 EXTENSION

Wordbank

algebraic expression An expression that describes a quantity by using variables and numerals, such as $3x + 4y - 8$

algebraic term A part of an algebraic expression, such as $3x$ in $4x^2 + 3x - 11$

evaluate To find the value of an expression

expand To rewrite an expression such as $4(2k + 4)$ without brackets

factorise The opposite of expand; to rewrite an expression, such as $8k + 16$, with brackets

like terms Terms that have exactly the same variables, for example, $5p$ and $2p$

substitute To replace a variable with a number

variable (or **pronumeral**) A quantity that can take on different values, represented by a symbol such as a letter of the alphabet

Quiz
Wordbank 3

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (19):

3.01 Algebraic expressions • Simplifying algebraic expressions 2
3.02 From words to algebraic expressions
3.03 Substitution 2 • Substitution 1
3.04 Adding and subtracting terms
3.05 Multiplying terms • Dividing terms
3.06 Powers, indices and exponents • Simplifying with the index laws • Index laws 1 • Index laws 2
3.07 Expanding expressions • Expanding expressions 2
3.09, 3.10 Factorising expressions
3.11 Expanding binomial products 1 • Expanding binomial products 2
3.12 Special binomial products
3.13 Difference of 2 squares • Special binomial products

Twig videos (2):

3.03 The heartbeat formula • The heptathlon

PhET interactives (2):

3.04 Expression exchange
3.07 Area model Algebra

Quizzes (6):

- Wordbank 3
- SkillCheck 3
- Mental skills 3A
- Mental skills 3B
- Language of maths 3
- Test yourself 3

Skillsheets (4):

3.01 Order of operations
3.02 Algebraic expressions
3.04, 3.07 Algebra using diagrams
3.08, 3.09, 3.10 Algebra using diagrams • Factorising using diagrams

Worksheets (14):

SkillCheck What does the symbol mean?
3.02 Writing algebraic expressions • What does the symbol mean?
3.03 Substitution
3.04 Collecting like terms
3.05 Simplifying algebraic expressions • Perimeter and area • Why aren't they the same?
3.07 Expanding brackets
3.10 Algebra review • Algebraic expressions review
3.11 Area diagrams • Binomial products • Special products
3.12, 3.13 Special products
Mind map Algebra

Puzzles (19):

3.01 Algebra bingo game
3.02 Generalised arithmetic • What's the expression?
3.03 Formula 1 game • Substitution game
3.04 Algebra bingo game • Adding and subtracting like terms
3.06 Indices puzzle
3.07 Expandominoes • Expanding • Expand the brackets • Expand negative brackets
3.09 Factorising
3.10 Expandominoes • Factorising puzzle • Factorising with negative terms
3.11 Expanding brackets • Trinomials
3.12 Factorominoes
3.13 Enough time
Language of maths Algebra find-a-word

Presentation (1):

3.04 Collecting like terms

Technology (1):

3.11 Expanding binomials

Spreadsheet (1):

3.11 Expanding binomials

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- learn about variables as a way of representing numbers using letters
- generalise number laws and properties to algebraic terms and expressions
- convert between word and algebraic expressions
- evaluate algebraic expressions by substituting values for each variable
- simplify algebraic expressions using the 4 arithmetic operations
- (EXTENSION, YEAR 9 CORE) apply index laws to algebraic expressions with whole number and zero powers
- use the distributive law to expand algebraic expressions with grouping symbols (brackets)
- factorise algebraic terms and expressions
- (EXTENSION, YEAR 9 PATH) expand binomial products, including perfect squares and difference of 2 squares

SkillCheck ANSWERS ON P. 575

Quiz
SkillCheck 3

1 Evaluate each expression.

a $3 - 4$ **b** $-5 + 9$ **c** $-8 - 2$

d $3 - (-2)$ **e** -3×6 **f** $-12 \div (-4)$

g $-12 \div 4 - 10$ **h** $-4 \times (-8)$ **i** $(-6)^2$

j 2^3 **k** $(-1)^3$ **l** 2×3^2

2 Simplify each algebraic expression.

a $7 \times p$ **b** $a \times b$ **c** $k \times k$

d $n \times n \times n$ **e** $2 \times p \times m$ **f** $p \times k \times d$

g $y \times 6$ **h** $t \times 3 \times q$ **i** $6d - 7d$

j $q + q + q$ **k** $a + a + a + a$ **l** $1x$

3 Choose which operation (+, −, × or ÷) is related to each word.

a sum **b** product **c** difference **d** quotient

e decrease **f** increase **g** more than **h** less than

4 If $x = 8$ and $y = 2$, evaluate each expression.

a $x + y$ **b** $3x$ **c** $x - y$ **d** $5y$

e xy **f** $9 - y$ **g** $x + 17$ **h** $2x + 4$

i $3y - 2$ **j** $x \div y$ **k** $5x \div y$ **l** $4x + 5y$

5 Find:

a the sum of 8 and 5 **b** double 9

c 6 increased by 10 **d** the difference between 14 and 11

e the product of 3 and 7 **f** how many times 5 divides into 40

Worksheet
What does the symbol mean?

3.01 Variables

Video
Algebraic expressions

Puzzle
Algebra bingo game

In algebra, a **variable** or **pronumeral** is a symbol, usually a letter of the alphabet, that stands for a number. It is called a variable because its value can vary (change).

When writing **algebraic expressions**, we use the following abbreviations.

- $3 \times k = 3k$ for multiplication, we leave out the '×' symbol
- $m \div 4 = \frac{m}{4}$ for division, we can write in fraction form
- $r \times r = r^2$ for powers
- $1x = x$ for multiplying by 1
- $c \times a \times 6 = 6ac$ write the number first, then the variables in alphabetical order

Video
Simplifying algebraic expressions 2

Example 1

Simplify each expression.

a $p + p + p + p$ **b** $d \times 5$ **c** $r \div 7$

d $m \times n \times 6$ **e** $4v + 2v$ **f** $5 \times k \times 2 \times h$

g $y \times y \times y$ **h** $5a - 4a$ **i** $b \times b \times 15$

Solution

a $p + p + p + p = 4p$ — 4 lots of p.

b $d \times 5 = 5d$ — Write the number first.

c $r \div 7 = \frac{r}{7}$

d $m \times n \times 6 = 6mn$ — Number first, then variables in alphabetical order.

e $4v + 2v = 6v$ — $(v + v + v + v) + (v + v) = 6$ lots of v

f $5 \times k \times 2 \times h = 5 \times 2 \times k \times h$
$= 10hk$

g $y \times y \times y = y^3$

h $5a - 4a = 1a$ — $(a + a + a + a + a) - (a + a + a + a) =$ one a left
$= a$ — Just a: no need to write the '1'.

i $b \times b \times 15 = 15b^2$ — The '2' (squared) belongs to the b only.

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Example 2

Use order of operations to simplify each expression.

a $2 \times m - 4$ **b** $15 - k \div 3$ **c** $13 \div (d \times d) + 4$

Solution

a $2 \times m - 4 = 2m - 4$ $\times$ first, then $-$

b $15 - k \div 3 = 15 - (k \div 3)$ $\div$ first, then $-$

$= 15 - \frac{k}{3}$

c $13 \div (d \times d) + 4 = 13 \div d^2 + 4$ ( ) first, then $\div$, then $+$

$= \frac{13}{d^2} + 4$

Skillsheet
Order of operations

3.01

Example 3

Write each expression in expanded form.

Note that expanding is the opposite of simplifying.

a $5bc$ **b** $-4kr^2$ **c** $x^2 - \frac{t}{11}$

Solution

a $5bc = 5 \times b \times c$ **b** $-4kr^2 = -4 \times k \times r \times r$ **c** $x^2 - \frac{t}{11} = x \times x - t \div 11$

EXERCISE 3.01 ANSWERS ON P. 575

Variables U F R C

EXAMPLE 1

1 Simplify each expression. R C

a $w \times 5$ **b** $m \times m$ **c** $3 \times c \times b \times a$

d $k + k + k + k + k + k$ **e** $m \div 8$ **f** $f + f + f$

g $12m + 4m$ **h** $7 \times w \times 3 \times x$ **i** $11p - 2p$

j $4 \times t \times t$ **k** $6d - 5d$ **l** $b \times c \times d \times 16$

m $26 \div n$ **n** $h \times 9 \times h$ **o** $9x + 11x$

p $a + a - a$ **q** $12q - 4q$ **r** $2 \times c \times 6 \times n$

s $3 \times r \times d \times (-2) \times d$ **t** $a + a + a + c + c$ **u** $a \times a \times a \times x \times x$

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EXAMPLE 2

2 Use order of operations to simplify each expression. R C

a $13 \div (a \times 2)$ **b** $9 + 4 \times m$ **c** $(e - 6) \div 5$

d $4 \times t - 8$ **e** $y \times y + z \times z$ **f** $k \div 9 - n$

g $p \div (9 + n)$ **h** $22 - e \div 2$ **i** $11 + u \times u \times 11$

3 Which expression is equal to $m \times m \times 5 + 2$? Select the correct answer **A**, **B**, **C** or **D**. R C

A $5m^2 + 2$ **B** $5m^3 + 2$ **C** $5m + 2$ **D** $m^2 + 7$

4 Explain the meaning of: C

a pq **b** $\frac{3y}{x}$ **c** $1x = x$

5 Which expression is **not** equal to $4p$? Select **A**, **B**, **C** or **D**. R C

A $5p - p$ **B** p^4 **C** $p + p + p + p$ **D** $p \times 4$

EXAMPLE 3

6 Write each expression in expanded form. C

a $8st$ **b** $-2y^2$ **c** $\frac{8}{f}$ **d** $r^2 + t^2$

e $4b^2 d$ **f** $5xy - 2a$ **g** $\frac{x-2}{3}$ **h** q^3

i $16 - 2d^2$ **j** $\frac{-4d}{3}$ **k** $h^2 i^2$ **l** $4j - \frac{9}{d}$

7 What is $7e^3$ in expanded form? Select **A**, **B**, **C** or **D**. C

A $7e \times 7e \times 7e$ **B** $7e + 7e + 7e$ **C** $7 \times e + e + e$ **D** $7 \times e \times e \times e$

Investigation

The laws of arithmetic

We can use algebraic symbols to describe general laws about arithmetic.

For example, if we add zero to any number, the answer is still that number.

If we let N stand for any number: $N + 0 = N$

In groups of 2 to 4, answer the following questions.

1 Describe in words what each number rule below means.

a $N \times 1 = N$ **b** $N \times 0 = 0$

c $a - b \neq b - a$ **d** $a + b + c = b + a + c$

e $N - 0 = N$ **f** $ab = ba$

2 Write each rule algebraically using variables.

a Any number divided by 1 equals itself.

b Multiplying a number by 8 is the same as doubling it 3 times.

c Any 3 numbers can be multiplied together in any order.

d Any number added to itself is the same as multiplying that number by 2.

e Any number subtracted from itself equals 0.

f Any number multiplied by its reciprocal equals 1.

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3 Is each equation true or false? Test your answer by substituting a number for each variable and checking.

a $a \div b = b \div a$

b $N \div N = 1$

c $4a - a = 4$

d a is a factor of a

e If N is even, then $N + 3$ is odd

f $\frac{1}{2}N = N - 2$

g $a + (-a) = 2a$

h $0 \div N = 0$

i $N \div 0 = 0$

j $N \div 1 = N$

4 If k is an odd number, what is an expression for:

a the previous odd number?

b the next even number?

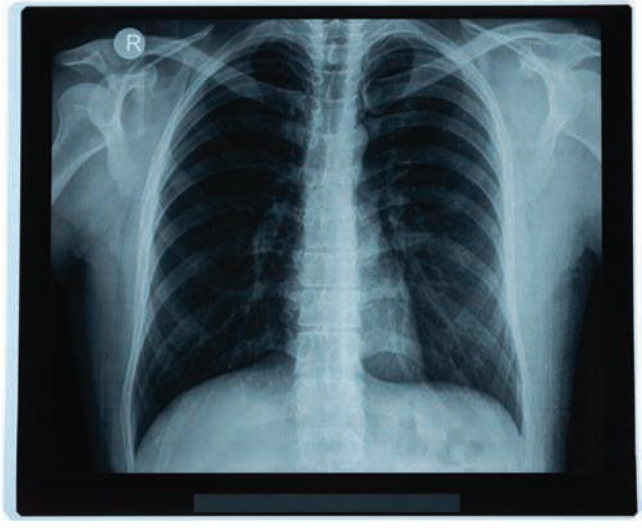
Did you know?

The X factor

In algebra the letter '***x***' is used quite frequently as a variable. During the 9th century, the Arabic word ***al-jabr*** ('algebra') described the process of moving variables from one side of an equation to the other to find the value of an unknown. The word for 'thing' or 'object' (the unknown number) in Arabic was ***shei***, translated into Greek as ***xei***, and shortened to ***x***. ***Xenos*** is also the Greek word for unknown, stranger or foreigner, which could explain why early European mathematicians used the letter ***x*** in algebra.

One further explanation is that ancient scholars could only write using feathers dipped in ink. The two easiest letters to write were ***x*** and ***y***. Mathematicians needed variables they could write quickly without mixing them up. As a result, x and y are still the two most commonly used variables in algebra.

Find examples of X-words that involve something unknown, such as X-ray, Brand X or X factor.



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3.02 Creating algebraic expressions

Video
From words to algebraic expressions

Skillsheet
Algebraic expressions

Worksheets
Writing algebraic expressions

What does the symbol mean?

Puzzles
Generalised arithmetic

What's the expression?

Example 4

If x is a variable representing any number, write an algebraic expression for:

a 4 times the number

b the number divided by 6

c the previous consecutive number

d one-third of the number

consecutive means 'following in order': for example, 7, 8, 9 are consecutive numbers

Solution

a $4 \times x = 4x$

b $x \div 6$ or $\frac{x}{6}$

c $x - 1$ The number before x is $x - 1$ (for example, the number before 4 is $4 - 1 = 3$).

d $\frac{1}{3}x = \frac{x}{3}$

Example 5

Think of a number, y, multiply it by 3 and subtract 7.
Write an algebraic expression for the answer.

Solution

First, we multiply the number y by 3 to get $y \times 3 = 3y$.

Then, we subtract 7 from our result to get the answer, $3y - 7$.

Example 6

Describe the algebraic expression $\frac{a+b}{2}$ in words.

Solution

$\frac{a+b}{2}$ is the sum of a and b, divided by 2, or the average of a and b.

EXERCISE 3.02 ANSWERS ON P. 576

Creating algebraic expressions U F R C

EXAMPLE 4

3.02

1 If n represents any number, write an algebraic expression for: R C

a 5 times the number
b the number plus 6
c the number divided by 10
d the next consecutive number
e half of the number
f 20 less than the number
g the number multiplied by itself
h 20 minus the number
i the previous consecutive number
j double the number
k 6 divided by the number
l the difference between the number and 8
m the product of the number and (-4)
n the square root of the number

2 If a, b and c represent any 3 numbers, write an algebraic expression for: R C

a the sum of a, b and c
b the product of a and b
c the difference between b and c, where b is greater than c
d the product of a and c, plus b
e 3 times b, plus a
f c divided by b
g how much a is greater than b
h 5 more than (b times c)
i c less than b
j a times b divided by c
k the square root of the sum of a and c
l the product of a and b, decreased by 7 times c

3 A cricketer scored a total of x runs in 6 innings. What was her average score? Select the correct answer **A, B, C** or **D**. R C

A $6x$
B $6 \div x$
C x^6
D $x \div 6$



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EXAMPLE 5

4 Write an algebraic expression for the answer to each instruction. C

a Think of a number (y), multiply it by 3 and add 8

b Think of a number (p), divide it by 3 and then subtract 6

c Think of a number (M), multiply it by 4 and then add 27

d Think of a number (k), multiply it by 5 and then divide it by 2

e Think of a number (x), divide it by 6 and add 13

f Think of a number (B), halve it and then subtract 2

g Think of a number (R), divide it by 9 and then add 10

h Think of a number (n), multiply it by 7 and then multiply it by 3

EXAMPLE 6

5 Describe each algebraic expression in words. C

a $x + y - z$ **b** $5K + L$ **c** $xy - 3$ **d** $\frac{W}{5}$

e $3x - 1$ **f** $\frac{y - x}{10}$ **g** $2N + 21$ **h** $7 - e$

6 Write an expression for: R C

a the number of girls in a class if there are b boys and a total of t students

b the number of bottles of drink needed for a party if each bottle can be shared by 5 guests and there are n guests

c the amount of change in dollars that Sandra receives if she paid for 3 hamburgers costing $\$p$ each with $\$y$ in cash

d the perimeter of an isosceles triangle with equal sides of length r and an uneven side of length 9

e the cost of one egg if a carton of a dozen eggs costs $\$a$

f the perimeter of a square of length x

g the number of rows of chairs needed to place 160 chairs in the hall if each row contains r chairs



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h the total cost of attending the pool for x adults and y children, where admission is \$5 per adult and \$3 per child

i the side length of an equilateral triangle with perimeter $5m - 1$

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Substitution 3.03

The word **substitute** means replacing one thing with another. In algebra, substitution means to replace a variable with a value to evaluate an algebraic expression.

Example 7

If $p = 3$ and $q = 14$, evaluate each expression.

a $2p + q$ **b** $\frac{pq+2}{4}$ **c** $5q - p^2$

Solution

a $2p+q=2\times3+14$
$=20$

b $\frac{pq+2}{4}=\frac{3\times14+2}{4}$
$=\frac{44}{4}$
$=11$

c $5q-p^2=5\times14-3^2$
$=61$

Substituting into formulas

A **formula** is a general rule that is written as an algebraic equation and shows the relationship between variables. Solving mathematical problems often involves substituting into formulas.

Example 8

The formula for the perimeter of a rectangle is $P = 2l + 2w$, where P is the perimeter, l is the length and w is the width. Use this formula to calculate the perimeter of a rectangle with length 12 m and width 9.5 m.

Solution

When length $l = 12$ and width $w = 9.5$,

$P=2l+2w$
$=2\times12+2\times9.5$
$=43$ m

EXERCISE 3.03 ANSWERS ON P. 576

Substitution U F PS

1 If $x = 5$ and $y = 6$, what is the value of $2x - y$? Select the correct answer **A**, **B**, **C** or **D**.

A 19 **B** -2 **C** 4 **D** 16

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Videos
Substitution 2
Substitution 1
The heartbeat formula
The heptathlon

Puzzles
Formula 1 game
Substitution game

Worksheet
Substitution

EXAMPLE 7

2 If $a = 3$, $b = 5$ and $c = 2$, evaluate each expression.

a	$a + b$	**b**	$4a$	**c**	c^2	**d**	$a + b + c$
e	abc	**f**	$3b - a$	**g**	$c + a - b$	**h**	$4a - 2c$
i	$10 - b$	**j**	$ab + c^2$	**k**	$c - b$	**l**	$\frac{2a}{c}$

3 If $d = 10$, $e = -3$ and $f = 5$, evaluate each expression.

a	def	**b**	f^2	**c**	e^3	**d**	$d \div f$
e	$de \div 2$	**f**	$d^2 - f^2$	**g**	$7e$	**h**	$3e + 2d - f$
i	$f + d - e$	**j**	$\frac{d}{2} - 3f$	**k**	$6ef$	**l**	$\frac{e+f}{4}$

4 If $p = -3$, what is the value of $2p^2$? Select **A**, **B**, **C** or **D**.

A	36	**B**	18	**C**	-18	**D**	-36

5 If $y = -1$ and $z = 4$, evaluate each expression.

a	$3 - 4z$	**b**	$-3y^2$	**c**	$5(z - 2)$	**d**	$y(z + 1)$
e	$\frac{y}{z}$	**f**	$\frac{z}{4}$	**g**	$\frac{y+z}{9}$	**h**	$\frac{10}{3+y}$

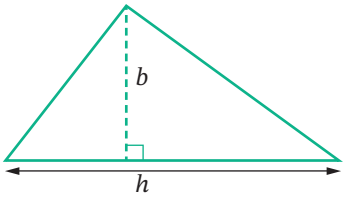
6 If $n = 6$, which of the following expressions is equal to 22? Select **A**, **B**, **C** or **D**.

A	$\frac{n+4}{2}$	**B**	$\frac{2n+4}{2}$	**C**	$\frac{n^2}{2}+4$	**D**	$\frac{n^2+4}{2}$

EXAMPLE 8

7 The formula for the area of a triangle is $A = \frac{1}{2}bh$, where A is the area of the triangle, b is the length of the base and h is the height.

Find the area of a triangle with base length 12 m and height 7 m.



8 A plumber charges according to the formula $C = 42h + 65$, where C is the charge in dollars and h is the number of hours worked. How much does the plumber charge for working for $3\frac{1}{2}$ hours?



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9 The formula for converting Celsius temperatures to Fahrenheit temperatures is $F = \frac{9}{5}C + 32$, where F is the temperature in Fahrenheit and C is the temperature in Celsius. Use the formula to convert 26°C to Fahrenheit. PS

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10 The number of hours of sleep recommended for children has the formula $H = 17 - \frac{A}{2}$, where H is the number of hours and A is the age of the child.

Find the number of hours of sleep recommended for a 6-year-old. PS

11 The formula for the cost of a party is $C = 45n + 500$, where C is the cost in dollars and n is the number of guests. Use this formula to calculate the cost of a party for 120 guests. PS

12 The time (t minutes) taken to travel to school depends on how far the student walks and how far the student travels on the bus. The formula for the time taken is $t = 11w + 4b$, where w is the walking distance and b is the distance travelled by bus.

a Find the time taken for each student to travel to school: PS

i Michael, who walks 1 km and goes 5 km on the bus

ii Vittoria, who walks 4 km and goes 3 km on the bus

b If school starts at 9 a.m., what is the latest time Wendy can leave home if she needs to walk 2 km and go on the bus for 7 km and still reach school on time?

Technology

Substitution

1 Enter the variables and values shown into a new spreadsheet.

2 In cell D1, enter the formula **=B4-B9** to evaluate $p - u$. You should get 9 because $4 - (-5) = 9$.

3 In cell D2, enter the formula **=B2+B3** to evaluate $m + n$. You should get 22 because $10 + 12 = 22$.

4 For each cell below, enter a formula to evaluate the given expression.

	A	B
1	variable	value
2	m	10
3	n	12
4	p	4
5	q	0
6	r	2.7
7	s	-8
8	t	25
9	u	-5

This means m = **10.**

D3: pr

D4: $\frac{t}{u}$

D5: $n + t - p$

D6: $n - \frac{m}{u}$

D7: $\frac{n-m}{u}$

D8: $\frac{s}{p} \times r$

D9: p^3

D10: $(s + n) \times (p + r)$

D11: $t + pq + s$

D12: $\sqrt{t}$

D13: $mnpq$

D14: the average of r, s, t and u

D15: $\frac{6m-n}{p+s}$

D16: $\sqrt{n+p}$

D17: ts^2

D18: $n^3 - r^2$

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3.04 Adding and subtracting terms

Skillsheet
Algebra using diagrams

Worksheet
Collecting like terms

Puzzles
Algebra bingo game

Adding and subtracting like terms

Interactive
Expression exchange

Suppose an unknown number p is represented by a cup and another unknown number m is represented by an envelope. Now consider these 2 sums.

a



$p + m + p + m = 2p + 2m$

b



$m + p + m + p + m = 3m + 2p$

Note that we can add all the p **terms** together, and we can add all the m terms together, but we cannot add the ps and ms together because p and m represent different numbers. We cannot simplify the answers $2p + 2m$ and $3m + 2p$ because p and m are different.

These are examples of collecting **like terms**, terms that contain exactly the same variable(s), such as $4h$, $5h$ and h. Here are some more examples.

$3x$ and $-5x$ $\quad$ $4mw$ and $2mw$ $\quad$ $12m$ and m

$-6d$ and $10d$ $\quad$ $5ab$ and $2ba$ $\quad$ xyz and $2yzx$

Note: $ab = ba$

Here are some examples of **unlike terms**, terms that have different variables.

$3x$ and $5m$ $\quad$ $4mw$ and $2mn$ $\quad$ $-6d$ and $10d^2$

Adding and subtracting terms

When **adding and subtracting terms**, only **like terms** can be collected.

Like terms have exactly the same variables.

Example 9

Select the like terms from each list below.

a $5x, -x, x^2, xy, \frac{1}{2}x, -8x, \frac{4x}{3}$ $\quad$ **b** $3ab, 5, a, ba, -2b, 4ab^2, -ab$

Solution

a $5x, -x, \frac{1}{2}x, -8x, \frac{4x}{3}$ are like terms because they each have a single x in them.

b $3ab$, ba and $-ab$ are like terms because each contains ab.

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Example 10

Simplify each expression by collecting like terms.

a $7d + 3e + 3d + e$

b $6m - 8 + 3m + 6$

c $6xy + 7x^2 - 3yx - 12x^2$

d $14 - 4t - 9t + 2$

Solution

a $7d + 3e + 3d + e = 7d + 3d + 3e + e$

$= 10d + 4e$

Grouping like terms

b $6m - 8 + 3m + 6 = 6m + 3m - 8 + 6$

$= 9m - 2$

The + or − sign goes with the term after it.

c $6xy + 7x^2 - 3yx - 12x^2 = 6xy - 3xy + 7x^2 - 12x^2$

$= 3xy - 5x^2$

$3yx = 3xy$

d $14 - 4t - 9t + 2 = 14 + 2 - 4t - 9t$

$= 16 - 13t$

Video
Adding and subtracting terms

Presentation
Collecting like terms

3.04

EXERCISE 3.04 ANSWERS ON P. 576

Adding and subtracting terms U F R C

EXAMPLE 9

1 Select the like terms from each list. C

a $4p, 2y, 2p, 5z, p$

b $m^2, m, n, 3m^2, mn$

c $5ac, 4a, 7ca, 5$

d $vw, 5v, 9v, w, v^2, 2wv$

e $p, 6pq, 2qp, 3pq, 7q$

f $2d, 3d^3, 7d^2, 9, 3d$

g $4mn, 3m, 4, 2nm, mn, 2n$

h $x^2y, 2x, 3y, 4x^2y$

i $7, 2a, 4b, 5ba, ab$

2 What are the like terms in $3ab, 2a^2, 2b^2a, 2p^2, 2ba$?
Select the correct answer **A**, **B**, **C** or **D**. C

A $2a^2, 2p^2$

B $2a^2, 2b^2a, 2p^2, 2ba$

C $3ab, 2ba$

D $2b^2a, 2ba$

3 Simplify each expression by collecting like terms.

a $4k + 7k$

b $5mn + 2nm$

c $xy + xy$

d $3abc + 4abc + 2bac$

e $3x^2 + 2x^2 + x^2$

f $ef^2 + 4ef^2$

g $8d - 3d$

h $12mk - 7mk$

i $12xy - 7yx - xy$

j $8de - 12de$

k $6k^2 - 3k^2$

l $4w^2 - w^2$

EXAMPLE 10

4 Which expression is $2m + 4p - 5m - 2p$ simplified? Select **A**, **B**, **C** or **D**.

A $-mp$

B $7m + 6p$

C $-3m + 2p$

D $-3p$

Foundation Standard Complex

5 Simplify each expression.

a	$5k - 2j + 6j$	**b**	$10ab + 2 - ab$	**c**	$12m^2 - 5m^2 + 2m$
d	$12s^2 - 7st - 2s^2$	**e**	$4mn + 7mn - mn$	**f**	$14k + 3 - 5k$
g	$y^2 + 2y^2 - y^2$	**h**	$9d - 4e + 3d$	**i**	$7 - 2x - 3$
j	$15gb - 8gb + gb$	**k**	$4 - 2q + 5$	**l**	$5a + 6c - 4c$

6 Simplify each expression.

a	$4x - y - x - 2y$	**b**	$7ab + 2bc - 3ab + bc$	**c**	$4k^2 + 3 + 5k^2 + 8$
d	$4bc - 8a + 2bc - a$	**e**	$x^2 - 6x - x + 7$	**f**	$p^2 + q^2 + 3p^2 - q^2$
g	$3 + 4y + 8y + 11$	**h**	$11r - 3s - 4r + 6s$	**i**	$7g + 8b - b - g$
j	$6x^2 - 4x - 9x + 6$	**k**	$20a + 11m + 32a - 11m$	**l**	$3q - 7 - 8q + 12q$
m	$5 - 2y - 5 - 4y$	**n**	$s^2 + 4s^2 + 12 - 5s^2$		

7 Which expression is $a^5 + a^5$ simplified? Select **A**, **B**, **C** or **D**.

A a^{10} **B** a^{25} **C** $2a^5$ **D** $2a^{10}$

8 Write an algebraic expression involving a sum that simplifies to: R C

a	$2x + 6$	**b**	$6m + 12p$	**c**	$x^2 - x$

9 Write an algebraic expression involving a difference that simplifies to: R C

a	$2x + 6$	**b**	$6m + 12p$	**c**	$x^2 - x$

3.05 Multiplying and dividing terms

Worksheets
Why aren't they the same?
Simplifying algebraic expressions
Perimeter and area

The associative law of multiplication tells us that to calculate the product $6 \times 4 \times 5 \times 3$ mentally, we can rearrange the numbers to pair convenient factors like this:

$$\begin{aligned} 6 \times 4 \times 5 \times 3 &= (4 \times 5) \times (6 \times 3) \\ &= 20 \times 18 \\ &= 360 \end{aligned}$$

We can do the same algebraically. As numbers can be multiplied in any order, we can multiply like *and* unlike terms.

Video
Multiplying terms

Example 11

Simplify each product.

a	$7b \times 3a$	**b**	$8x \times (-2y)$	**c**	$5d \times 6d \times 4$
d	$9m^2 \times 4p$	**e**	$4a \times 3b \times (-2a)$		

Foundation Standard Complex

Solution

a $7b \times 3a = 7 \times b \times 3 \times a$

$= 7 \times 3 \times a \times b$ Group numbers and variables separately

$= 21ab$

b $8x \times (-2y) = 8 \times (-2) \times x \times y$

$= -16xy$

c $5d \times 6d \times 4 = 5 \times 6 \times 4 \times d \times d$

$= 120d^2$

d $9m^2 \times 4p = 9 \times 4 \times m^2 \times p$

$= 36m^2p$

e $4a \times 3b \times (-2a) = 4 \times 3 \times (-2) \times a \times a \times b$

$= -24a^2b$

3.05

Multiplying terms

When **multiplying algebraic terms**, multiply the numbers first, then multiply the variables. Write all variables in alphabetical order.

Example 12

Use index notation to simplify each expression.

a $k \times k \times k \times k$

b $j \times j \times j \times j \times j \times j$

Video
Multiplying terms

Solution

a $k \times k \times k \times k = k^4$

b $j \times j \times j \times j \times j \times j = j^6$

Dividing terms

When **dividing algebraic terms**, divide the numbers first, then divide the variables. Write all variables in alphabetical order.

Example 13

Simplify each quotient.

a $20p \div 5$

b $\frac{24y}{8y}$

c $\frac{-2m}{4mn}$

d $12a^2b \div 3a$

Video
Dividing terms

Solution

a $20p \div 5 = \left(\frac{20}{5}\right)p$

$= 4p$

b $\frac{24y}{8y} = \left(\frac{24}{8}\right)\frac{y}{y}$

$= 3$

c $\frac{-2m}{4mn} = \left(\frac{-2}{4}\right)\frac{m}{mn}$

$= \left(-\frac{1}{2}\right)\frac{1}{n}$

$= -\frac{1}{2n}$

d $12a^2b \div 3a = \left(\frac{12}{3}\right)\frac{aab}{a}$

$= 4ab$

EXERCISE 3.05 ANSWERS ON P. 576

Multiplying and dividing terms UFRC

EXAMPLE 11

1 Simplify $2p \times 5pq$. Select the correct answer **A**, **B**, **C** or **D**.

A $10pq$ **B** $7p^2q$ **C** $7pq$ **D** $10p^2q$

2 Simplify each expression.

a $5 \times 3y$ **b** $8 \times 2m$ **c** $4k \times m$ **d** $4t \times 5$

e $2 \times f \times 6$ **f** $2x \times 8y$ **g** $10p \times 3m$ **h** $3b \times 6d$

i $3d \times 4d$ **j** $5 \times 4m \times 6n$ **k** $2y \times 3y \times 8$ **l** $3r \times 3 \times 3r$

m $2 \times 4a \times 9b$ **n** $x \times xy \times y$ **o** $2j \times 3k \times 7$ **p** $5rst \times 2rs$

q $3ac \times 6cd$ **r** $hjk \times hj \times jk$ **s** $4q^2 \times 6p^2$ **t** $2m \times 3n \times mn$

3 Simplify each expression.

a $-4b \times 2d$ **b** $6m \times (-4m)$ **c** $-2 \times 5k \times 9l$ **d** $20 \times (-3p)$

e $9k \times (-7)$ **f** $-15b \times (-5b)$ **g** $-3m \times mn \times 4n$ **h** $6abc \times (-2ab)$

i $5r \times (-2r) \times (-4)$ **j** $9d \times (-4e) \times 6d$ **k** $-8x \times 9x \times (-2)$ **l** $5m \times 4 \times (-2m)$

4 Which expression simplifies to $14a^2n^2$? Select **A**, **B**, **C** or **D**.

A $2a \times an \times 7$ **B** $2a \times 7n \times an$ **C** $7a^2 \times 2n$ **D** $7a^2n^2 \times 2a^2n^2$

EXAMPLE 12

5 Simplify each expression.

a $y \times y \times y \times y$ **b** $a \times a \times a$ **c** $2 \times c \times c$

d $t \times t \times t \times t \times t$ **e** $q \times q \times q \times 7$ **f** $k \times k \times k \times l \times l$

g $m \times m \times m \times n \times n$ **h** $-5 \times p \times p \times p \times p \times k \times k \times k$

i $-3 \times y \times y \times y$ **j** $d \times d \times (-1) \times e \times e \times f$ **k** $q \times q \times r \times r \times r \times r \times 4$

EXAMPLE 13

6 Simplify $\frac{55m^2n}{5m}$. Select **A**, **B**, **C** or **D**.

A $11mn$ **B** $11m^2n$ **C** $50mn$ **D** $11n$

7 Simplify each quotient.

a $12m \div 3$ **b** $36e \div 4$ **c** $21p \div (-3)$ **d** $\frac{81x}{9}$

e $\frac{60w}{15}$ **f** $\frac{8m}{m}$ **g** $-25k \div 5k$ **h** $30y \div 40y$

i $28pq \div 14q$ **j** $\frac{63a}{-9a}$ **k** $\frac{4pq}{20q}$ **l** $\frac{-56mn}{8m^2}$

8 Simplify each quotient.

a $14m^2 \div 7$ **b** $6e^2 \div (-2e)$ **c** $-5p^2 \div p$ **d** $\frac{14x^2}{2x}$

e $\frac{6wm}{-3m}$ **f** $\frac{8m^2n}{2mn}$ **g** $32k^2y \div (-4k)$ **h** $9jk^2 \div 12jk^2$

i $\frac{15d^2e^2}{3de}$ **j** $\frac{3a^2b^2c}{3ab^2c}$ **k** $\frac{-24acb}{3bc}$ **l** $\frac{-10m^2n}{5mn}$

Foundation Standard Complex

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9 Simplify $12x \div (-3) \times 2x$. Select **A**, **B**, **C** or **D**.

A $-2x$ **B** -2 **C** $-8x$ **D** $-8x^2$

10 Simplify each expression.

a $3p + 2 \times 4p$ **b** $7a^2 - 2a \times 3a$ **c** $6r \times 2y - y \times 10r$

d $5p - 4p \times 8$ **e** $8x \times 9x \div 12$ **f** $48m^2 \div 12m \times 2n$

g $6x + 4x \div 2$ **h** $24p - \frac{18p^2}{3p}$ **i** $6w^2 - \frac{10w^5}{5w^3}$

j $x^4 - \frac{x^7}{x^3}$ **k** $\frac{15mn^2p}{3m^2n^2}$ **l** $\frac{-4x^2y \times 3y}{2xy \times (-6xy)}$

11 Write an algebraic expression involving a product that simplifies to: R C

a $18de$ **b** $-20a^2y$ **c** $5pr^2$

12 Write an algebraic expression involving a quotient that simplifies to: R C

a $7m$ **b** $-3ab$ **c** $5pr^2$

3.05

Quiz
Mental skills 3A

Mental skills 3A: Maths without calculators ANSWERS ON P. 577

Simplifying multiplication by factorising

Sometimes, one big multiplication may be made into many simpler little multiplications if we break one or both terms into 2 'easier' factors. Then we use the property that numbers can be multiplied in any order.

1 Study each example.

a $24 \times 8 = 6 \times 4 \times 2 \times 4$
$= 4 \times 4 \times 6 \times 2$
$= 16 \times 6 \times 2$
$= 96 \times 2$
$= 192$

Look for pairs of numbers that you can multiply easily and rearrange

b $15 \times 6 = 3 \times 5 \times 6$
$= 3 \times 30$
$= 90$

c $14 \times 36 = 7 \times 2 \times 6 \times 6$
$= 6 \times 7 \times 6 \times 2$
$= 42 \times 6 \times 2$
$= 252 \times 2$
$= 504$

d $30 \times 22 = 10 \times 3 \times 22$
$= 10 \times 66$
$= 660$

Note: For each question, there are many different ways of arriving at the correct result.

2 Now evaluate each product by factorising first.

a 30×24 **b** 18×27 **c** 25×33 **d** 16×8

e 28×20 **f** 12×18 **g** 21×9 **h** 36×12

i 16×35 **j** 22×28 **k** 9×15 **l** 27×25

Foundation Standard Complex

Investigation

Equivalent expressions

State whether each statement below is true or false, then check by substituting a number for the variable and testing whether the left-hand side equals the right-hand side.

a $6 + 4m + 4 = 4m + 10$ **b** $5x + 3x = 8x^2$ **c** $4k - 7k = -3k$

d $7g + 8h - h - g = 7h + 6g$ **e** $3 + 7n + 13 + 2n = 25n$ **f** $3p \times 7q = 21pq$

g $21m \div 3m = 7m$ **h** $5y - 4y = y$ **i** $\dfrac{-16a}{4} = -4a$

3.06 Extension: The index laws

YEAR 9 CORE

Puzzle
Indices puzzle

Videos
Powers, indices and exponents
Simplifying with the index laws
Index laws 1
Index laws 2

In the previous chapter, *Working with numbers*, we discovered properties about powers, called **index laws**.

Here are some examples:

- $2^3 \times 2^4 = 2^{3+4} = 2^7$
- $7^5 \times 7 = 7^{5+1} = 7^6$
- $4^6 \div 4^2 = 4^{6-2} = 4^4$
- $\dfrac{10^3}{10^2} = 10^{3-2} = 10^1$
- $(3^4)^2 = 3^{4\times 2} = 3^8$
- $(8^5)^3 = 8^{5\times 3} = 8^{15}$
- $5^0 = 1$
- $8^0 = 1$

Now we can write these index laws algebraically and use them to simplify algebraic expressions.

Index laws

When **multiplying terms with powers** of the same base, **add** the powers.

$$a^m \times a^n = a^{m+n}$$

When **dividing terms with powers** of the same base, **subtract** the powers.

$$a^m \div a^n = \frac{a^m}{a^n} = a^{m-n}$$

When **raising a term with a power** to another power, **multiply** the powers.

$$(a^m)^n = a^{m\times n}$$

Any number raised to the **power of zero** equals **1**.

$$a^0 = 1$$

Now we can apply these index laws algebraically to variables.

Example 14

Simplify each expression.

a $p^6 \times p^2$ **b** $3m^2n \times 7m^3n^5$ **c** $a^{12} \div a^4$ **d** $\dfrac{42y^6}{7y^2}$

e $(d^4)^5$ **f** $(4m^2)^3$ **g** $2k^0$ **h** $(2k)^0$

Solution

a $p^6 \times p^2 = p^{6+2}$ Adding the powers
$= p^8$

b $3m^2n \times 7m^3n^5 = 3 \times 7 \times m^{2+3} \times n^{1+5}$
$= 21m^5n^6$

c $a^{12} \div a^4 = a^{12-4}$ Subtracting the powers
$= a^8$

d $\dfrac{42y^6}{7y^2} = \dfrac{42}{7} \times y^{6-2}$
$= 6y^4$

e $(d^4)^5 = d^{4\times5} = d^{20}$ Multiplying the powers

f $(4m^2)^3 = 4^3 \times (m^2)^3$ because $(4m^2)^3 = 4m^2 \times 4m^2 \times 4m^2$
$= 64m^{2\times3}$
$= 64m^6$

g $2k^0 = 2 \times k^0$ The zero power belongs to k only
$= 2 \times 1$
$= 2$

h $(2k)^0 = 1$ The zero power belongs to $(2k)$

EXERCISE 3.06 ANSWERS ON P. 577

The index laws U F

1 Simplify $a^2b \times a^4b^3$. Select the correct answer **A**, **B**, **C** or **D**.

A a^8b^3 **B** a^8b^4 **C** a^6b^3 **D** a^6b^4

EXAMPLE 14

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YEAR 9 CORE

2 Simplify each product.

a	$m^5 \times m^2$	**b**	$k^8 \times k$	**c**	$y^4 \times y^6$	**d**	$x^7 \times x$
e	$n^4 \times n^4$	**f**	$q^2 \times q^8$	**g**	$b^4 \times b^{10} \times b^2$	**h**	$s^3 \times s^5 \times s^7$
i	$6k^3 \times 7k^3$	**j**	$5y^3 \times 4y$	**k**	$d^4 \times 3d^5$	**l**	$11e^3 \times 5e$
m	$h^4n^2 \times h^6n^5$	**n**	$10c^3d^4 \times (-3c^3d)$	**o**	$f^5g^4 \times f^3g^7$	**p**	$-5p^3q \times 2p^4q$

3 Simplify $\frac{36d^6e^4}{12d^3e^4}$. Select **A**, **B**, **C** or **D**.

A	$3d^3$	**B**	$3d^3e$	**C**	$3d^2e$	**D**	$3d^2$

4 Simplify each quotient.

a	$y^6 \div y^5$	**b**	$p^4 \div p$	**c**	$\frac{q^5}{q^2}$	**d**	$n^8 \div n^3$
e	$\frac{k^{11}}{k^9}$	**f**	$x^7 \div x^5$	**g**	$r^8 \div r^8$	**h**	$m^{12} \div m^5 \div m^2$
i	$q^5 \times q^3 \div q^7$	**j**	$45g^6 \div 5g^2$	**k**	$18p^7 \div 6p^3$	**l**	$a^{12} \div a^6 \div a$
m	$66x^4y^2 \div 6x^3y$	**n**	$\frac{x^4 \times x^2}{x^3}$	**o**	$\frac{70k^5l^6}{10k^2l}$	**p**	$\frac{15e^3f}{21ef^2}$

5 Simplify $(a^4b^2)^3$. Select **A**, **B**, **C** or **D**.

A	a^7b^5	**B**	$a^{12}b^6$	**C**	$a^{12}b^5$	**D**	a^7b^6

6 Simplify each expression.

a	$(y^8)^2$	**b**	$(x^2)^5$	**c**	$(y^3)^3$	**d**	$(m^2)^9$
e	$(k^7)^3$	**f**	$(t^{20})^2$	**g**	$(3m)^2$	**h**	$(2m^4)^3$
i	$(xy)^3$	**j**	$(p^3q)^4$	**k**	$(3h^4)^4$	**l**	$(k^3z)^5$
m	$(2kj)^4$	**n**	$(4c^3d)^2$	**o**	$(5x^2y^8)^3$	**p**	$(2m^4p^2)^6$

7 Simplify each expression.

a	y^0	**b**	$(xy)^0$	**c**	$(2r - r)^0$	**d**	$5u^0$
e	$\left(\frac{x}{2}\right)^0$	**f**	$(2k^7)^0$	**g**	$m^8 \div m^8$	**h**	$h^2 \div (h^0)^3$
i	$(b^3)^2 \div 2b^0$	**j**	$(3x)^0 \times (4x)^2$	**k**	$(d^0)^5 \times (d^8)^0$	**l**	$26x^4y^3 \div 2x^3y^3$
m	$4m^0 + 3m - 3k^0$	**n**	$\frac{27b^3d}{9bd}$	**o**	$(8h)^2 \times 2h^0 \div 4h^2$	**p**	$\frac{6ef^4}{12ef^4}$

8 Simplify each expression.

a $\frac{3pq}{9p^2q^2}$ **b** $\frac{2m^3n^3}{m^2n^2}$ **c** $\frac{8xy}{12xy^2}$

Foundation Standard Complex

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Expanding expressions

3.07

The **distributive law** says that you can multiply by a number by splitting it into the sum or difference of 2 other numbers. Look at these 2 examples.

a $23 \times 12 = 23 \times (10 + 2)$

$= (10 + 2) + (10 + 2) + (10 + 2) + \ldots$ 23 times

$= 10+10+10+\ldots+2+2+2+\ldots$ 23 times each

$= (23\times10)+(23\times2)$

$= 230+46$

$= 276$

so $23 \times (10 + 2) = 23 \times 10 + 23 \times 2$

b $35 \times 9 = 35 \times (10 - 1)$

$= (10 - 1) + (10 - 1) + (10 - 1) + \ldots$ 35 times

$= 10+10+10+\ldots-1-1-1-$ 35 times each

$= (35\times10)-(35\times1)$

$= 350-35$

$= 315$

so $35 \times (10 - 1) = 35 \times 10 - 35 \times 1$

Skillsheet Algebra using diagrams

3.07

Worksheet Expanding brackets

Puzzles Expandominoes

Expanding

Expand the brackets

Expand negative brackets

Interactive Area model Algebra

The distributive law for expanding algebraic expressions

If a, b and c stand for numbers, then: $a(b + c)$ means $a \times (b + c)$

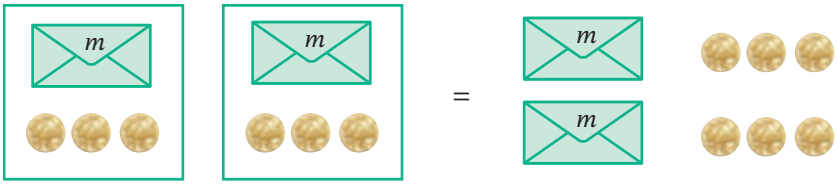
$$a(b + c) = ab + ac$$

$$a(b - c) = ab - ac$$

We can apply the distributive law to algebraic expressions. Suppose an unknown number m is represented by an envelope 

that contains m coins.

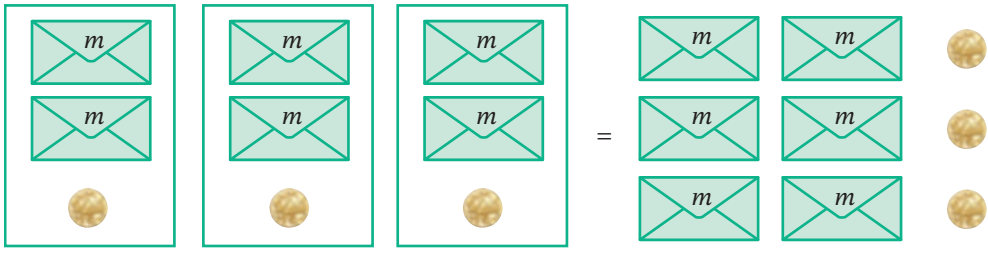
Then $2(m + 3)$ or '2 lots of $(m + 3)$', can be represented by this diagram.



It can be seen that $2(m + 3) = 2 \times m + 2 \times 3 = 2m + 6$.

The number outside the brackets is *multiplied* by each term inside.

Here is another example for $3(2m + 1)$.



It can be seen that $3(2m + 1) = 3 \times 2m + 3 \times 1 = 6m + 3$.

Removing the brackets in an algebraic expression by multiplying each term inside by the term outside is called **expanding** the expression.

Video
Expanding expressions

Example 15

Expand each expression.

a $7(p + 4)$ **b** $3(x - 5)$ **c** $5(2m - 7)$

d $-2(a + 3)$ **e** $-(4d - 2)$ **f** $w(9w - 1)$

Solution

a $7(p + 4) = 7 \times p + 7 \times 4$
$= 7p + 28$

7 is multiplied by each term inside the brackets separately.

b $3(x - 5) = 3 \times x - 3 \times 5$
$= 3x - 15$

c $5(2m - 7) = 5 \times 2m - 5 \times 7$
$= 10m - 35$

d $-2(a + 3) = -2 \times a + (-2) \times 3$
$= -2a + (-6)$
$= -2a - 6$

-2 is multiplied by each term inside the brackets separately.

e $-(4d - 2) = -1(4d - 2)$
$= -1 \times 4d - (-1) \times 2$
$= -4d + 2$

$-(4d - 2)$ is the same as $-1 \times (4d - 2)$.

f $w(9w - 1) = w \times 9w - w \times 1$
$= 9w^2 - w$

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Example 16

Expand and simplify each expression.

a $2(x+5)+6(x-4)$ **b** $9k-4(k-3)$ **c** $3p-(p+6)$

Solution

a $2(x+5)+6(x-4)=2\times x+2\times 5+6\times x+6\times(-4)$ Expanding

$=2x+10+6x-24$

$=8x-14$ Collecting like terms

b $9k-4(k-3)=9k-4\times k-4\times(-3)$

$=9k-4k+12$

$=5k+12$

c $3p-(p+6)=3p-1(p+6)$

$=3p-1\times p-1\times 6$

$=3p-p-6$

$=2p-6$

Video
Expanding expressions 2

3.07

EXAMPLE 15

EXERCISE 3.07 ANSWERS ON P. 577

Expanding expressions U F R C

1 Expand each expression.

a $5(m+3)$	**b** $8(m-2)$	**c** $2(j+10)$	**d** $3(a+2)$
e $4(h-7)$	**f** $6(x+y)$	**g** $9(4-b)$	**h** $11(y-6)$
i $4(2k+5)$	**j** $3(5m-2)$	**k** $6(3x+4)$	**l** $10(2a+5b)$

2 Expand each expression.

a $-3(n+4)$	**b** $-4(v-5)$	**c** $-7(d+3)$	**d** $-10(k-2)$
e $-2(d+e)$	**f** $-5(5-z)$	**g** $-6(h+1)$	**h** $-(m-1)$
i $-9(5+2x)$	**j** $-(2f+3)$	**k** $-3(u-v)$	**l** $-8(4-k)$

3 Expand $-3(a-2b)$. Select the correct answer **A**, **B**, **C** or **D**.

A $-3a-2b$ **B** $-3a+6b$ **C** $-3a-6b$ **D** $-3a+5b$

4 Expand each expression.

a $x(y+6)$	**b** $p(q-2)$	**c** $a(a+4)$	**d** $b(3b-1)$
e $d(d+3e)$	**f** $2(4m+5n)$	**g** $5k(3k-2)$	**h** $7j(3j+ik)$
i $x(4x+y)$	**j** $-2(4p-3q)$	**k** $k(k-12)$	**l** $-m(4-m)$
m $4n(2n+5m)$	**n** $2x(1-7x)$	**o** $-f(2f+4)$	**p** $-(5j-7k)$

5 Expand $3y(y-4)$. Select **A**, **B**, **C** or **D**.

A $3y+12$ **B** $3y^2-7y$ **C** $3y^2-12y$ **D** $3y+7$

6 Explain the difference between $2a+1$ and $2(a+1)$. R C

7 Scott says that, to find the perimeter of a rectangle, you double its length, double its width, and add the answers together. Thomas says that you add the length and width first, then double the answer. Who is correct? Can you write each of their answers algebraically? R C

Foundation Standard Complex

EXAMPLE 16

8 Expand $12 - 8(x + 2)$. Select **A**, **B**, **C** or **D**.

A $4(x + 2)$ **B** $4x + 8$ **C** $-8x + 14$ **D** $-8x - 4$

9 Expand and simplify each expression.

a $6(a + 2) + 14$ **b** $4k + 2(k + 3)$ **c** $3(q - 3m) + 5m$

d $8 + 6(p - 5)$ **e** $18 - (2y + 14)$ **f** $31 - 7(t + 2)$

g $6(2h - 1) + 11$ **h** $4x - 4(3x - 2)$ **i** $2(9n - 2) + 8$

10 Check that $8(2b - 3) = 16b - 24$ by substituting any value for b and showing that the answers to both sides are equal. R

11 Expand and simplify each expression.

a $3(a + 4) + 2(a + 1)$ **b** $2(x + 5) + 6(x - 1)$ **c** $7(4 + k) + 5(2k - 2)$

d $5(2t + 1) + 3(4t + 2)$ **e** $3(n + 5) - 2(n + 4)$ **f** $10(d + 2) - 6(d + 3)$

g $8(4y + 1) - 5(2y - 3)$ **h** $3(p - 4) - 5(p - 1)$ **i** $m(m + 1) + 3(m + 1)$

j $4(d + e) + 3(d - e)$ **k** $2f(3f + 4) - 4(2f - 1)$ **l** $5e(1 - 7e) - e(3e + 4)$

12 Copy and complete each equation. R

a $2(_____) = 2x + 10$ **b** $4(_____) = 4r - 12$ **c** $3(_____) = 24k - 9$

d $5(_____) = 10a + 35$ **e** $-2(_____) = -2y - 18$ **f** $-3(_____) = -6t + 21$

3.08 Factorising algebraic terms

Skillsheets
Algebra using diagrams

Factorising using diagrams

The **factors** of a number are those numbers that divide into it evenly.
For example, the factors of 12 are 1, 2, 3, 4, 6 and 12.

Algebraic terms also have factors. The **factors** of an algebraic term are those numbers or terms that divide into it evenly. For example, $10xy = 2 \times 5 \times x \times y$, so *some* of the factors of $10xy$ are 2, 5, x, y, 10, $2y$ and $10x$. Often there are too many to list them all easily.

Example 17

List 4 factors of $40a^2b$.

Solution

There are many ways of multiplying terms to get $40a^2b$, such as:

$8 \times 5 \times a \times a \times b$

$4 \times 10 \times a^2 \times b$

$40 \times a^2 \times b$

So 4 factors of $40a^2b$ are: 4, 5, a and a^2b. (*Note*: It is impractical to list them all.)

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The highest common factor of algebraic terms

The **highest common factor (HCF)** (or **greatest common divisor (GCD)**) of 2 numbers is the largest number that divides into both numbers evenly. For example, the HCF of 18 and 30 is 6.

The **highest common factor** of 2 algebraic terms is the largest term that divides into both terms evenly. For example, the HCF of $18ab$ and $30b$ is $6b$.

Highest common factor

To find the **highest common factor (HCF)** of 2 or more algebraic terms:

- find the HCF of the numbers
- find the HCF of the variables
- multiply them together to make the highest common algebraic factor.

Example 18

Find the highest common factor of each pair of terms.

a $16x$ and $40xy$ **b** $12r$ and 28

Solution

a First, find the HCF of the numbers.

The factors of 16 are **1**, **2**, **4**, **8**, 16.

The factors of 40 are **1**, **2**, **4**, 5, **8**, 10, 20, 40.

So the HCF of 16 and 40 is **8**.

Then find the HCF of the variables:

The HCF of x and xy is x.

$\therefore$ The HCF of $16x$ and $40xy$ is $8 \times x = 8x$.

b First, find the HCF of the numbers:

The factors of 12 are **1**, **2**, 3, **4**, 6, 12.

The factors of 28 are **1**, **2**, **4**, 7, 14, 28.

So the HCF of 12 and 28 is **4**.

Then find the HCF of the variables:

The second term (28) has no variable so the HCF of the variables is 1.

$\therefore$ The HCF of $12r$ and 28 is $4 \times 1 = 4$.

EXERCISE 3.08 ANSWERS ON P. 577

Factorising algebraic terms U F R

1 List the factors of each number.

a 15 **b** 8 **c** 17

d 25 **e** 55 **f** 42

2 Find the highest common factor of each pair of numbers. R

a 9 and 6 **b** 6 and 14 **c** 18 and 24

d 16 and 12 **e** 33 and 22 **f** 15 and 25

g 21 and 9 **h** 45 and 30 **i** 16 and 48

EXAMPLE 17

3 List 4 factors of each term.

a $5m$ **b** $7pq$ **c** $10x$ **d** $18d$

e $4k^2$ **f** $21bc$ **g** $32x^2y$ **h** $15ab$

4 Which term is **NOT** a factor of $36m^2n$? Select the correct answer **A**, **B**, **C** or **D**.

A $4m$ **B** $9n$ **C** $8mn$ **D** $12m^2n$

EXAMPLE 18

5 Find the highest common factor of each pair of terms. R

a $3xy$ and $12x$ **b** $4j$ and $24j$ **c** $6x$ and $4ax$

d $21kj$ and $14k$ **e** 5 and $10p$ **f** $16m$ and $12m^2$

g $48cd$ and $32bc$ **h** de and de^2 **i** 9 and $27x^2$

j $3m$ and $6n$ **k** $22m$ and $4mn$ **l** $16y$ and $4y$

m a^2x and ay **n** $64j$ and 32 **o** $15w^2$ and $20wy$

p $22x^2y$ and $36xy^2$ **q** cd and ad **r** $36t$ and $12ty^2$

6 Which term is the highest common factor of $8x^2$ and $12xy$? Select **A**, **B**, **C** or **D**.

A 4 **B** $2x$ **C** $4x$ **D** $24x^2y$

3.09 Factorising expressions

Skillsheets
Algebra using diagrams

Factorising using diagrams

Expanding an algebraic expression means removing its brackets by multiplying terms.

Factorising is the opposite of expanding. To factorise an algebraic expression, find the HCF of all terms and insert brackets.

expanding

$3(x + 6) = 3x + 18$

factorising

Factorising algebraic expressions

Puzzle
Factorising

- Find the HCF of all terms and write it outside the brackets
- Divide each term by the HCF and write them inside the brackets

$ab + ac = a(b + c)$

$ab - ac = a(b - c)$

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To check that the answer is correct, expand it.

Example 19

Factorise each expression.

a $2x + 6$ **b** $7 - 7y$ **c** $3ab - 5b$ **d** $r^2 + 7rs$

Solution

a $2x + 6 = 2 \times x + 2 \times 3$
$= 2(x + 3)$

HCF of $2x$ and 6 is 2.
Think: $2 \times$ ___ $= 2x$, $2 \times$ ___ $= 6$
Check by expanding: $2(x + 3) = 2x + 6$

b $7 - 7y = 7 \times 1 - 7 \times y$
$= 7(1 - y)$

HCF of 7 and $7y$ is 7.
Think: $7 \times$ ___ $= 7$, $7 \times$ ___ $= 7y$
Check by expanding: $7(1 - y) = 7 - 7y$

c $3ab - 5b = b \times 3a - b \times 5$
$= b(3a - 5)$

HCF of $3ab$ and $5b$ is b.
Think: $b \times$ ___ $= 3ab$, $b \times$ ___ $= 5b$
Check by expanding: $b(3a - 5) = 3ab - 5b$

d $r^2 + 7rs = r \times r + r \times 7s$
$= r(r + 7s)$

HCF of r^2 and $7rs$ is r.
Think: $r \times$ ___ $= r^2$, $r \times$ ___ $= 7rs$
Check by expanding: $r(r + 7s) = r^2 + 7rs$

Video
Factorising expressions

3.09

EXERCISE 3.09 ANSWERS ON P. 577

Factorising expressions U F R C

1 Find the highest common factor of the terms in each expression. R C

a $10a + 2b$ **b** $4 - 12x$ **c** $ax + xy$ **d** $y^2 - y$

2 Copy and complete each factorisation. R C

a $2x + 4 = 2($_____$)$ **b** $4p - 16 = 4($_____$)$
c $12k - 15 = 3($_____$)$ **d** $5j - 20k = 5($_____$)$
e $3x^2 + 3y^2 = 3($_____$)$ **f** $5x - 15 =$ _____$(x - 3)$
g $12x - 6 =$ _____$(2x - 1)$ **h** $2a + 8b =$ _____$(a + 4b)$
i $4pq + 10 =$ _____$(2pq + 5)$ **j** $5 + 15x = 5($_____$)$
k $xy + xz = x($_____$)$ **l** $ab - ac = a($_____$)$
m $pq - p = p($_____$)$ **n** $cd + bc = c($_____$)$
o $a^2 + 2a = a($_____$)$ **p** $5hk - 8h =$ _____$(5k - 8)$
q $24mn + 11n =$ _____$(24m + 11)$ **r** $xy + y^2 =$ _____$(x + y)$
s $3k^2 + 4jk =$ _____$(3k + 4j)$ **t** $ef - 2f^2 =$ _____$(e - 2f)$

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EXAMPLE 19

3 Factorise each expression. R C

a	$4x + 8$	**b**	$3m + 6$	**c**	$5a - 10$	**d**	$15m - 10a$
e	$8x - 4y$	**f**	$12m - 16w$	**g**	$hk - h$	**h**	$2xy - xm$
i	$12p + 5pr$	**j**	$x^2 + 5x$	**k**	$4m - m^2$	**l**	$y^2 - 11y$

4 Factorise $16k^2 + 18k$. Select the correct answer **A**, **B**, **C** or **D**.

A $2(8k^2 + 9k)$ **B** $k(16k + 18)$ **C** $2k(8k + 9)$ **D** $4k(4k + 9)$

5 Copy and complete each factorisation. R C

a	$2am + 2an = 2a($_____$)$	**b**	$6xy - 3x = 3x($_____$)$
c	$4ap + 2a = 2a($_____$)$	**d**	$5mn - 5mp = 5m($_____$)$
e	$3xy - 12x = 3x($_____$)$	**f**	$2pq + 2pr = 2p($_____$)$
g	$14x - 2xy = 2x($_____$)$	**h**	$20kj + 5j = 5j($_____$)$
i	$9xy - 12yz = 3y($_____$)$		

6 Factorise $3d^2 + 21d$. Select **A**, **B**, **C** or **D**.

A $3(d^2 + 7d)$ **B** $d(3d + 21)$ **C** $d(3d + 7)$ **D** $3d(d + 7)$

7 Factorise each expression. R C

a	$7ab + 14a$	**b**	$2x - 6xy$	**c**	$3m - 9mw$	**d**	$10xy - 25xw$
e	$8am - 12mp$	**f**	$9xy + 6my$	**g**	$3x + 9xy$	**h**	$ab + abc$
i	$xyz + xya$	**j**	$18m - 24mn$	**k**	$2ac + 3abc$	**l**	$pqr + mpq$
m	$rs + 7r$	**n**	$4s^2 + 16$	**o**	$p^2 - 2p$	**p**	$6k^3 - 2k$
q	$3j + j^2$	**r**	$7n - n^2$	**s**	$7e + 21e^2$	**t**	$6q - 12q^2$
u	$a^4 + a^2$	**v**	$x^2y + xy^2$	**w**	$3m^2n + 9m$	**x**	$12a^2b^2c^2 + 3abc$

3.10 Factorising with negative terms

Skillsheets
Algebra using diagrams

Factorising using diagrams

Worksheets
Algebra review

Algebraic expressions review

Factorising with negative terms

When factorising an algebraic expression whose first term is *negative*, include the negative sign in the factor.

$$-ab - ac = -a(b + c)$$
$$-ab + ac = -a(b - c)$$

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Example 20

Factorise each expression.

a $-5x - 10$ **b** $-9n + 21$ **c** $-gh - h$ **d** $-mn + n^2$

Solution

a $-5x - 10 = -5 \times x + (-5) \times 2$
$= -5(x + 2)$

HCF of $-5x$ and -10 is -5.
Think: $-5 \times ___ = -5x$, $-5 \times ___ = -10$
Check by expanding: $-5(x + 2) = -5x - 10$

b $-9n + 21 = -3 \times 3n + (-3) \times (-7)$
$= -3(3n + (-7))$
$= -3(3n - 7)$

HCF of $-9n$ and 21 is -3.
Think: $-3 \times ___ = -9n$, $-3 \times ___ = 21$
Check by expanding: $-3(3n - 7) = -9n + 21$

c $-gh - h = -h \times g + (-h) \times 1$
$= -h(g + 1)$

HCF of $-gh$ and $-h$ is $-h$.
Think: $-h \times ___ = -gh$, $-h \times ___ = -h$
Check by expanding: $-h(g + 1) = -gh - h$

d $-mn + n^2 = -n \times m + (-n) \times (-n)$
$= -n(m + (-n))$
$= -n(m - n)$

HCF of $-mn$ and n^2 is $-n$.
Think: $-n \times ___ = -mn$, $-n \times ___ = n^2$
Check by expanding: $-n(m - n) = -mn + n^2$

Video
Factorising expressions

Puzzles
Expandominoes
Factorising puzzle
Factorising with negative terms

EXERCISE 3.10 ANSWERS ON P. 577

Factorising with negative terms U F R C

1 Copy and complete each factorisation. R C

a $-6k - 6 = -6(_____)$ **b** $-mn - mp = -m(_____)$
c $-5r + 10 = -5(_____)$ **d** $-3k + 27 = -3(_____)$
e $-ac + ad = _____(c - d)$ **f** $-y^2 - 7y = _____(y + 7)$
g $-12y - 30 = _____(2y + 5)$ **h** $-4f + f^2 = _____(4 - f)$

2 Factorise each expression. R C

EXAMPLE 20

a $-4x + 12$ **b** $-3m - 9$ **c** $-10x + 20$ **d** $-15x + 10m$
e $-6x - 8m$ **f** $-5x - 30$ **g** $-a - ab$ **h** $-4xyz + 8abc$
i $-20 + 4x$ **j** $-dx - dy$ **k** $-2m + 5mp$ **l** $-24x - 16y$

3 Factorise $-14wy - 28wz$. Select the correct answer **A**, **B**, **C** or **D**.

A $-2(7wy + 14wz)$ **B** $-14w(y + 2z)$ **C** $14w(-y - 2z)$ **D** $-2w(7y + 14z)$

4 Factorise each expression. R C

a $-16xy - 24xk$ **b** $-20x + 12xy$ **c** $-8mw + 12m$
d $5e - 15ef$ **e** $-ab - abc$ **f** $xya + xyw$
g $-12c - 18cw$ **h** $-2rt + 7rst$ **i** $20fh + 5h$
j $-19bw - 38w$ **k** $-abc + abcd$ **l** $-abc - abcd$

Foundation Standard Complex

5 **a** Substitute $w = 2$ into the expressions $-w^2 + 7w$ and $-w(w - 7)$.

b Is $-w^2 + 7w = -w(w - 7)$ for any value of w? R C

6 Factorise $-ab + a^2$. Select **A**, **B**, **C** or **D**.

A $-a(b - a)$ **B** $a(-b - a)$ **C** $-a(b + a)$ **D** $-a(b + a^2)$

7 Factorise each expression. R C

a $a^4bc - a^3b^2c$ **b** $2\pi r^2 + 2\pi rh$

Quiz
Mental skills 3B

Mental skills 3B: Maths without calculators ANSWERS ON P. 578

Multiplying by 9, 11, 99 or 101

We can use expanding when multiplying by a number near 10 or near 100.

1 Study each example.

a
$$\begin{aligned} 25 \times 11 &= 25 \times (10 + 1) \\ &= 25 \times 10 + 25 \times 1 \\ &= 250 + 25 \\ &= 275 \end{aligned}$$

b
$$\begin{aligned} 14 \times 9 &= 14 \times (10 - 1) \\ &= 14 \times 10 - 14 \times 1 \\ &= 140 - 14 \\ &= 126 \end{aligned}$$

c
$$\begin{aligned} 32 \times 12 &= 32 \times (10 + 2) \\ &= 32 \times 10 + 32 \times 2 \\ &= 320 + 64 \\ &= 384 \end{aligned}$$

d
$$\begin{aligned} 7 \times 99 &= 7 \times (100 - 1) \\ &= 7 \times 100 - 7 \times 1 \\ &= 700 - 7 \\ &= 693 \end{aligned}$$

e
$$\begin{aligned} 27 \times 101 &= 27 \times (100 + 1) \\ &= 27 \times 100 + 27 \times 1 \\ &= 2700 + 27 \\ &= 2727 \end{aligned}$$

f
$$\begin{aligned} 18 \times 8 &= 18 \times (10 - 2) \\ &= 18 \times 10 - 18 \times 2 \\ &= 180 - 36 \\ &= 144 \end{aligned}$$

2 Now evaluate each product.

a	16×11	**b**	33×11	**c**	29×9	**d**	45×9
e	62×11	**f**	7×101	**g**	18×101	**h**	36×99
i	19×8	**j**	45×12	**k**	21×102	**l**	6×98

3.11 Extension: Expanding binomial products

YEAR 9 CORE

$(x + 4)$ and $(x - 7)$ are called **binomial expressions** because they each have exactly 2 terms.

$(x + 3)(x - 1)$ is called a **binomial product** because it is a product of 2 binomial expressions.

binomial = '2 terms'

Foundation Standard Complex

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Worksheets
Area diagrams
Binomial products
Special products

Puzzles
Expanding brackets
Trinomials

Technology
Expanding binomials

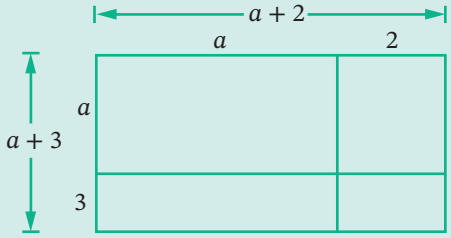
Spreadsheet
Expanding binomials

Example 21

Expand $(a + 2)(a + 3)$ using an area diagram.

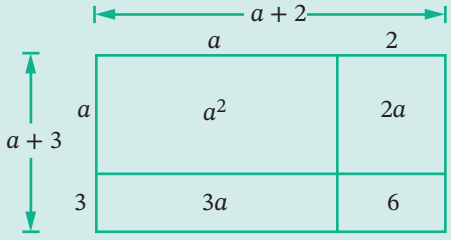
Solution

Draw an area diagram (rectangle) with length $(a + 2)$ and width $(a + 3)$ and divide the diagram into 4 smaller rectangles.



Find the area of each smaller rectangle.

Expand $(a + 2)(a + 3)$ by adding the areas of the 4 rectangles.



$(a + 2)(a + 3) = a^2 + 2a + 3a + 6$ Expanding

$= a^2 + 5a + 6$ Simplifying by collecting like terms.

Another way of expanding binomial products is to use the distributive law. Each term in the first binomial is multiplied by each term in the second binomial to give **4 terms**, which are collected and simplified.

Example 22

Expand each binomial product.

a $(x + 5)(x + 9)$ **b** $(k + 3)(k - 7)$ **c** $(x - 6)(4x + 2)$

d $(3t - 1)(2t - 5)$ **e** $(a - 6)^2$

Solution

a $(x + 5)(x + 9) = x(x + 9) + 5(x + 9)$

$= x^2 + 9x + 5x + 45$ Expanding to make 4 terms.

$= x^2 + 14x + 45$ Simplifying.

b $(k + 3)(k - 7) = k(k - 7) + 3(k - 7)$

$= k^2 - 7k + 3k - 21$

$= k^2 - 4k - 21$

c $(x - 6)(4x + 2) = x(4x + 2) - 6(4x + 2)$

$= 4x^2 + 2x - 24x - 12$

$= 4x^2 - 22x - 12$

d $(3t - 1)(2t - 5) = 3t(2t - 5) - 1(2t - 5)$

$= 6t^2 - 15t - 2t + 5$

$= 6t^2 - 17t + 5$

e $(a - 6)^2 = (a - 6)(a - 6)$

$= a(a - 6) - 6(a - 6)$

$= a^2 - 6a - 6a + 36$

$= a^2 - 12a + 36$

Videos
Expanding binomial products 1
Expanding binomial products 2

YEAR 9 CORE

EXERCISE 3.11 ANSWERS ON P. 578

Expanding binomial products UFRC

EXAMPLE 21

1 Expand each binomial product by copying and completing the area diagram.

a $(x + 2)(x + 7)$

	x	2
x	---	---
7	---	---

b $(x + 5)(x + 1)$

	x	5
x	---	---
1	---	---

2 Expand $(p + 6)(p + 2)$. Select the correct answer **A**, **B**, **C** or **D**.

A $p^2 + 12$ **B** $p^2 + 6p + 12$ **C** $p^2 + 8p + 8$ **D** $p^2 + 8p + 12$

EXAMPLE 22

3 Use the distributive law to expand each binomial product algebraically. R

a $(x + 1)(x + 4)$ **b** $(x + 6)(x + 3)$ **c** $(x + 5)(x + 4)$ **d** $(x + 2)(x + 2)$

e $(x + 2)(x + 6)$ **f** $(x + 10)(x + 3)$ **g** $(x + 8)(x + 1)$ **h** $(x + 7)(x + 7)$

i $(x - 1)(x + 4)$ **j** $(x - 3)(x + 2)$ **k** $(x + 1)(x - 4)$ **l** $(x - 5)(x + 6)$

m $(x - 4)(x - 5)$ **n** $(x + 9)(x - 3)$ **o** $(x - 5)(x - 3)$ **p** $(x - 7)(x - 7)$

4 **a** Expand each binomial product.

i $(x + 3)(x - 3)$ **ii** $(x + 4)(x - 4)$ **iii** $(x - 2)(x + 2)$ **iv** $(x - 7)(x + 7)$

b Can you see a simpler way of expanding these products? Explain how you would expand $(x + 8)(x - 8)$. R C

5 Expand each binomial product.

a $(x + 1)(2x + 4)$ **b** $(3x - 1)(x + 2)$ **c** $(2x + 3)(x + 1)$ **d** $(x - 3)(5x + 2)$

e $(4x - 3)(x - 2)$ **f** $(5x + 2)(2x - 3)$ **g** $(x - 3)(3x - 1)$ **h** $(4x + 7)(x - 3)$

6 Expand each binomial square.

a $(a + 5)^2$ **b** $(x + 2)^2$ **c** $(y + 7)^2$

d $(x - 3)^2$ **e** $(2a + 1)^2$ **f** $(3a - 5)^2$

7 Expand:

a $(a + b)^2$ **b** $(a + b)(a - b)$

8 Expand $\left(4x + \frac{1}{2}\right)^2$. Select **A**, **B**, **C** or **D**.

A $16x^2 + 4x + \frac{1}{4}$ **B** $16x^2 + \frac{1}{4}$ **C** $4x^2 + 2x + \frac{1}{4}$ **D** $16x^2 + 2x + \frac{1}{4}$

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Extension: Perfect squares

3.12

Another name for **square number** is **perfect square**.

9 is a **perfect square** because it is 3^2.

36 is another **perfect square** because it is 6^2.

There is a special pattern when you expand a binomial multiplied by itself, for example, $(x + 5)(x + 5)$.

$(x + 5)(x + 5)$ is a perfect square because it is $(x + 5)^2$.

$(x + 5)^2$ is also called **the square of a binomial**, or a **perfect square**.

YEAR 9 PATH

Puzzle
Factorominoes

Worksheet
Special products

Example 23

Expand each perfect square.

a $(x + 5)^2$ **b** $(3p + 2)^2$

Solution

a $(x + 5)^2 = (x + 5)(x + 5)$

$= x(x + 5) + 5(x + 5)$

$= x^2 + 5x + 5x + 25$

$= x^2 + 10x + 25$

b $(3p + 2)^2 = (3p + 2)(3p + 2)$

$= 3p(3p + 2) + 2(3p + 2)$

$= 9p^2 + 6p + 6p + 4$

$= 9p^2 + 12p + 4$

Do you notice the pattern between the question and the answer for these perfect squares?

Perfect squares

The formulas for expanding the perfect square of a binomial are:

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

Proof 1: Using an area diagram

	a	b
a	a^2	ab
b	ab	b^2

$(a + b)^2 = a^2 + ab + ab + b^2$

$= a^2 + 2ab + b^2$

Proof 2: Using the distributive law

$(a + b)^2 = (a + b)(a + b)$

$= a(a + b) + b(a + b)$

$= a^2 + ab + ab + b^2$

$= a^2 + 2ab + b^2$

YEAR 9 PATH

Video
Special binomial products

Example 24

Expand each perfect square.

a $(k + 4)^2$ **b** $(3y - 8)^2$

Solution

a $(k + 4)^2 = k^2 + 2 \times k \times 4 + 4^2$
$= k^2 + 8k + 16$

b $(3y - 8)^2 = (3y)^2 - 2 \times (3y) \times 8 + 8^2$
$= 9y^2 - 48y + 64$

EXERCISE 3.12 ANSWERS ON P. 578

Perfect squares U F R

EXAMPLE 23

1 Find the missing number in each expansion.

a $(x + 6)^2 = x^2 + 12x + \square$ **b** $(x - 3)^2 = x^2 - \square x + 9$

c $(m - 4)^2 = m^2 - \square m + 16$

2 Expand $(x - 9)^2$. Select the correct answer **A**, **B**, **C** or **D**.

A $x^2 - 9x + 81$ **B** $x^2 - 18x - 81$ **C** $x^2 + 18x + 81$ **D** $x^2 - 18x + 81$

EXAMPLE 24

3 Expand each perfect square.

a $(y + 6)^2$ **b** $(x - 1)^2$ **c** $(h + 3)^2$ **d** $(n - 8)^2$

e $(a - 7)^2$ **f** $(x + 9)^2$ **g** $(y - 11)^2$ **h** $(d + 10)^2$

4 Expand each perfect square.

a $(2a + 3)^2$ **b** $(3x + 5)^2$ **c** $(2a - 1)^2$ **d** $(4x + 3)^2$

e $(5y - 1)^2$ **f** $(2p + 7)^2$ **g** $(3y - 5)^2$ **h** $(3 - 3a)^2$

5 Use the expansion of a perfect square to evaluate each square number without using a calculator. R

a $11^2 = (10 + 1)^2$ **b** $19^2 = (20 - 1)^2$ **c** $51^2 = (50 + 1)^2$

d $35^2 = (30 + 5)^2$ **e** $65^2 = (60 + 5)^2$ **f** $82^2 = (80 + 2)^2$

6 Expand and simplify each expression.

a $(a + 3)^2 + 4$ **b** $(x - 5)^2 - 3x$ **c** $2 + (a - 1)^2$ **d** $(x - 3)^2 - 3(x - 3)$

Foundation Standard Complex

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Extension: Difference of 2 squares

3.13

There is also a special pattern when you multiply the sum of 2 terms by the difference of those 2 terms, for example, $(x + 7)(x - 7)$.

$$\begin{aligned}(x + 3)(x - 3) &= x(x - 3) + 3(x - 3)\\ &= x^2 - 3x + 3x - 9\\ &= x^2 - 9\end{aligned}$$

Notice that the 2 middle terms cancel each other out and you are left with the 1st term squared minus the 2nd term squared. The answer is called the **difference of 2 squares**.

Difference of 2 squares

$$(a + b)(a - b) = a^2 - b^2$$

Example 25

Expand each expression.

a $(d + 3)(d - 3)$ **b** $(7x + 2)(7x - 2)$

Solution

a $(d + 3)(d - 3) = d^2 - 3^2$
$= d^2 - 9$

b $(7x + 2)(7x - 2) = (7x)^2 - 2^2$
$= 49x^2 - 4$

EXERCISE 3.13 ANSWERS ON P. 578

Difference of 2 squares U F R C

1 Expand each expression.

a $(x + 3)(x - 3)$ **b** $(y - 4)(y + 4)$ **c** $(x - 6)(x + 6)$ **d** $(d + 2)(d - 2)$
e $(x - 1)(x + 1)$ **f** $(n - 8)(n + 8)$ **g** $(k + 11)(k - 11)$ **h** $(a - 12)(a + 12)$

2 Expand each expression.

a $(2x + 1)(2x - 1)$ **b** $(3y - 5)(3y + 5)$ **c** $(5x - 6)(5x + 6)$ **d** $(3d + 5)(3d - 5)$
e $(4x - 1)(4x + 1)$ **f** $(6n - 8)(6n + 8)$ **g** $(2k + 7)(2k - 7)$ **h** $(6a - 2)(6a + 2)$

YEAR 9 PATH

3.13

Worksheet
Special products

Puzzle
Enough time

Videos
Difference of 2 squares

Special binomial products

EXAMPLE 25

Foundation Standard Complex

YEAR 9 PATH

3 Expand each expression.

a $(1 + x)(1 - x)$ **b** $(3 - m)(3 + m)$ **c** $(5 - 6a)(5 + 6a)$ **d** $(2d + 3)(2d - 3)$

e $(3x - m)(3x + m)$ **f** $(2n - 5k)(2n + 5k)$ **g** $(k + 7d)(k - 7d)$ **h** $(3a - 5c)(3a + 5c)$

4 By expressing 101×99 as $(100 + 1)(100 - 1)$, use the difference of 2 squares to find the value of 101×99. R

5 Use the method of question 4 to evaluate each product. R C

a 51×49 **b** 81×79 **c** 29×31 **d** 98×102

Power plus ANSWERS ON P. 578

1 Simplify each expression.

a $-7p - 8q + 3p - 5q + 4q$ **b** $-m - 3n - 5m - 7mn$

c $3k - 2j - 5 - 4k + 6j + 9$ **d** $a^2 + 2b^2 + 3b^2 - a^2$

e $4x^2 + 6x - x^2 - 12x$ **f** $xy + yz + xz + 4xy + 3xz + yz$

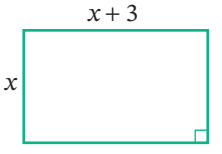
g $-d^2 + 7 + 6e^2 - 3d^2 - 2e^2 + 9$ **h** $5f^2 + 6f + 2 + 8f^2 - 1$

2 For each shape, write an algebraic expression for:

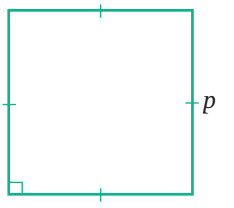
i its perimeter **ii** its area.

(Give your answer in simplest form.)

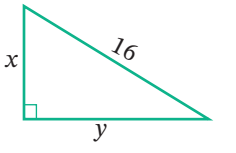
a



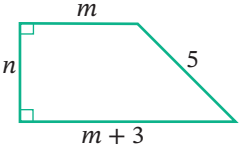
b



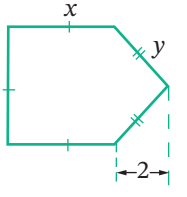
c



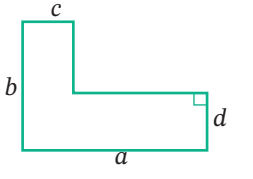
d



e



f



3 Expand and simplify each expression.

a $x(x - 3) + 5(x - 3)$ **b** $m(a + 4) + m(a - 7)$

c $-3(y - 4) + 7(y - 3)$ **d** $4(2a + b + 3c) - 2(a + 3b - c)$

e $p(p - 3q + 2r) + 2p(4r - 3p - 2q)$ **f** $7k^2 + 3(k^2 - 3k - 1) - (6k^2 - 4k + 2)$

g $12 - 5(4f - 6g - 3)$ **h** $5(2x^2 - 5x + 3) - 4(3x^2 - 8x - 25)$

i $(x + 3)(x + 5)$ **j** $(p - 1)(p + 10)$

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4 Evaluate each expression if $p = 4$, $q = -6$, $r = 3$ and $t = -1$.

a $p + q$ **b** pq **c** $r^2 + t^2$ **d** $q^2 - 6p + rt$

e $\frac{pq}{rt}$ **f** $6 - 2r + p$ **g** $2p + 3q + 4r - 5t$ **h** $\frac{p+4}{2}$

i $\frac{q+r}{p+t}$ **j** $\frac{r+7}{5} - \frac{q-t}{5}$ **k** $\frac{p+3}{10} + \frac{r-2}{2}$ **l** $q + pr - qt$

5 Factorise each expression.

a $9pq + 5p - 2p^2$ **b** $44k - 33l - 11m$ **c** $2x - 7x^2 + 3xy$

d $x(y + 2) + 3(y + 2)$ **e** $a(b - 3) - 7(b - 3)$ **f** $4(k + 1) - j(k + 1)$

6 Simplify each expression.

a $\frac{m}{2} + \frac{m}{2}$ **b** $\frac{7n}{10} - \frac{3n}{10}$ **c** $\frac{2k}{3} + \frac{k}{3}$ **d** $\frac{3d}{5} - \frac{d}{5}$

e $\frac{5f}{8} + \frac{7f}{8}$ **f** $\frac{3c}{10} + \frac{4c}{10}$ **g** $\frac{4}{x} + \frac{2}{x}$ **h** $\frac{7}{a} + \frac{5}{b}$

CHAPTER 3 REVIEW

Quiz
Language of maths 3

Puzzle
Algebra find-a-word

Language of maths

base	binomial	brackets	collecting
consecutive	distributive law	evaluate	expand
expanded form	expression	factor	factorise
formula	highest common factor (HCF)	index laws	indices
like terms	power	pronumeral	substitute
term	unknown	unlike terms	variable

1 Why does the word 'variable' have that name? Look up its meaning in the dictionary.

2 What are **unlike terms**?

3 How do you **expand** an algebraic expression?

4 What is the name given to the largest algebraic term that divides evenly into 2 or more given terms?

5 What is the opposite of **expanding** an algebraic expression?

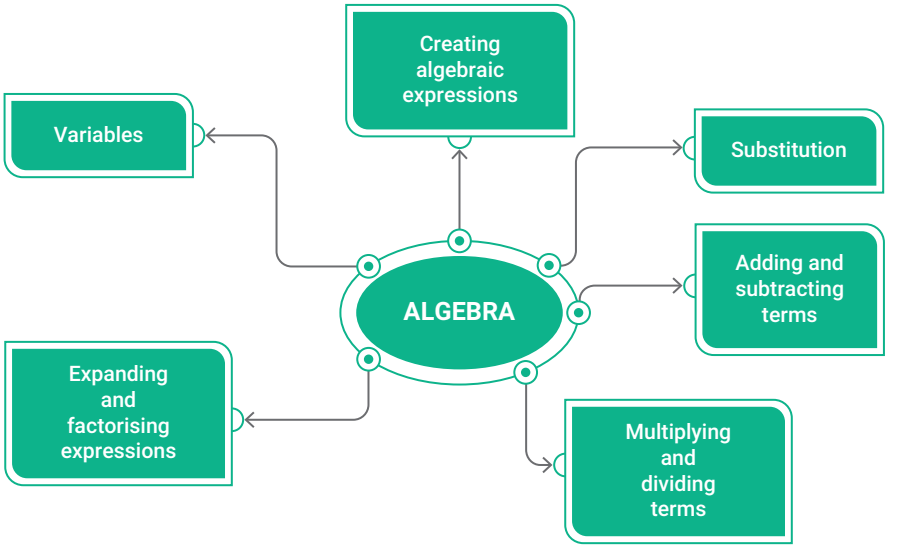
6 Look up some non-mathematical meanings of the word 'factor'. Use 'factor' in a sentence to show one of these meanings.

Worksheet
Mind map: Algebra

Topic summary

- What parts of algebra did you remember from Year 7? What new rules have you learnt?
- Are there parts of this chapter that you still do not understand? Talk to your teacher.
- Give 2 examples of jobs where algebra is needed.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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TEST YOURSELF 3 ANSWERS ON P. 578

Quiz
Test yourself 3

1 Simplify each expression. 3.01

a $p \times 6$ **b** $r \times (-1)$ **c** $4 \times d \times g$

d $t + t + t + t + t$ **e** $14m + 6m$ **f** $11k - 6k$

g $6 \times y \times y$ **h** $16 \div a$ **i** $5 \times c \times 4 \times d$

2 Write each expression in expanded form. 3.01

a cd **b** $3y^2$ **c** $\frac{w}{6}$ **d** $c^2 + y$

3 Write each statement as an algebraic expression. 3.02

a 4 more than y **b** 6 times b minus 3 **c** p divided by q

4 Evaluate $2n - p$ when $n = 11$ and $p = -3$. Select the correct answer **A**, **B**, **C** or **D**. 3.03

A 19 **B** 25 **C** 208 **D** 214

5 If $x = 5$ and $y = -2$, evaluate each expression. 3.03

a $x + y$ **b** xy **c** $3x - 2y$

d $6y^2$ **e** $6x \div y$ **f** $x^2 + y^2$

6 Simplify each expression. 3.04

a $5k + 7k$ **b** $29d - 12d$

c $4m + 18m - 5m$ **d** $13m - 3m + 8m$

e $4ac + 4ac - 5ca$ **f** $11cd + 7cd - 9cd$

g $7x + 3y + 3x + 7y$ **h** $13r + 4s + 11s + 2r$

i $33h + 6g - 5h + g$

7 Simplify each expression. 3.05

a $5 \times 2x$ **b** $2b \times 7d$ **c** $11m \times 6$

d $3p \times 8q$ **e** $4 \times 2w \times 4w$ **f** $15j \times 5$

g $3m \times 6m \times 2$ **h** $4p \times 3p \times 5p$

8 Simplify each expression. 3.05

a $50x \div 2$ **b** $28b \div 7d$ **c** $66m \div 6$

d $\frac{32rs}{4}$ **e** $\frac{15p}{3p}$ **f** $14d^2 \div 2$

g $14e^2 \div 2e$ **h** $13b^2 \div b$ **i** $\frac{18x^2}{9x}$

j $\frac{44m^2n^2}{11m}$ **k** $\frac{16e^2g}{4eg}$ **l** $\frac{35a^2e^2}{7ae}$

Foundation Standard Complex

3.07 **9** Expand and simplify each expression.

a $6(n + 2)$ **b** $-5(2b + 7)$

c $j(9j - 3)$ **d** $10 + 2(n + 4)$

e $22v - 3(v + 5)$ **f** $5(y + 2) + 4(y - 7)$

3.07 **10** Which expression is $5(c - d) - 2c + 3d$ simplified? Select **A**, **B**, **C** or **D**.

A $3c - 2d$ **B** $3c + 8d$ **C** $3c - 8d$ **D** $5c - 5d - 2c + 3d$

3.08 **11** Find the highest common factor of each pair of terms.

a $6a$ and 12 **b** $3p^2$ and $18p$ **c** $7t$ and 14

3.09 **12** Factorise each expression.

a $3m + 9$ **b** $5h - 35$ **c** $6a + 9ab$

d $10h - 12hm$ **e** $xy + xyz$ **f** $cd - de$

g $a^2 - 6a$ **h** $8kj + 24j^2$

3.10 **13** Factorise each expression.

a $-2 - 4q$ **b** $-4f - 16$ **c** $-12 + 15d$

d $-ab + b$ **e** $-16b - 24bd$ **f** $-4k^2 - k$

g $-x^2 + 7x$ **h** $-h^2i - hi^2$

YEAR 9 CORE

3.06 **14** Simplify each expression.

a $x^2 \times x^4$ **b** $m^6 \div m^2$ **c** $5k^2 \times 6k$ **d** $-2hy^7 \times 3h^2y^4$

e $n^7 \div n^3$ **f** $8a^4 \div 2a^3$ **g** $q^7 \div q^3 \div q$ **h** $\frac{9x^5y}{3x^2}$

i $\frac{28m^3n}{6mn^2}$ **j** $\left(w^3\right)^4$ **k** $\left(3a^4\right)^2$ **l** $\left(2d^3e^2\right)^5$

m p^0 **n** $6a^0$ **o** $5h^3 \div h^3$ **p** $(4x)^0 \times (3x)^2 \div 9x$

3.11 **15** Expand each binomial product.

a $(m + 9)(m + 2)$ **b** $(5h + 3)(h - 1)$ **c** $(y + 6)(y - 5)$ **d** $(2a - 3)(2a - 7)$

YEAR 9 PATH

3.12 **16** Expand each binomial product.

a $(m + 9)^2$ **b** $(h - 1)^2$ **c** $(2y - 5)^2$ **d** $(3a - 2)^2$

3.13 **17** Expand each binomial product.

a $(m + 9)(m - 9)$ **b** $(h - 3)(h + 3)$ **c** $(5y - 2)(5y + 2)$ **d** $(2a + 3c)(2a - 3c)$

Foundation Standard Complex

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PRACTICE SET 1

ANSWERS ON P. 579

1 Evaluate each expression. 2.08

a 7^3 **b** $(-35)^2$ **c** 4^4

d $\sqrt{400}$ **e** $\sqrt[3]{2744}$ **f** $\sqrt{2.25}$

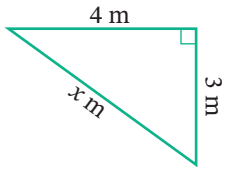
2 Evaluate each expression without using a calculator. 2.01

a 28×5 **b** 15×9 **c** $8 \times 3 \times 5$

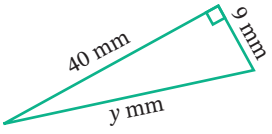
d 16×30 **e** $720 \div 4$ **f** $300 \div 20$

3 Find the value of each variable, correct to one decimal place where necessary. 1.05

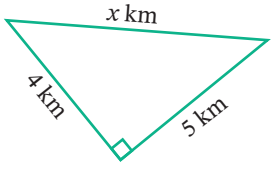
a



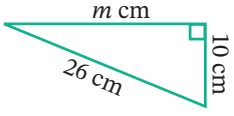
b



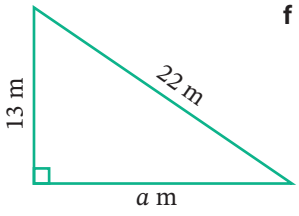
c



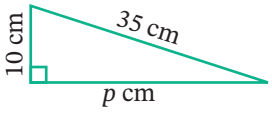
d



e



f



4 Evaluate each expression. 2.02 2.03

a $-3 + 11$ **b** $6 \times (-5)$ **c** $-11 - 9$

d $3 - (-2)$ **e** $-7 \times (-4)$ **f** $\frac{42}{-7}$

g $7 - 9$ **h** $-4 - (-6)$ **i** $-6 - 10$

j -4×7 **k** $-54 \div (-9)$ **l** $75 \div (-5)$

5 Evaluate each expression. 2.04

a $4 + 7 \times 2$ **b** $16 - 28 \div (-4)$

c $(-32 + 11) \div 7$ **d** $32 \div (-8) - 7 \times 1$

e $(3 + 7 \times 5) \div 2$ **f** $(17 - 6) \times (3 - 12)$

g $\frac{12 + 7}{4}$ **h** $\frac{8.8}{6.5 - 5.4}$

i $\frac{64 - 12}{15 + 11}$

6 If $a = 3$ and $b = 5$, then evaluate each expression. 3.03

a $2b + a$ **b** a^2 **c** $5a - b$ **d** $ab - b$

Foundation Standard Complex

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3.04

7 Simplify each expression.

a $3k + 2j - k + 2j$ **b** $5p - p$

c $4p + 3q + 7p + 2q$ **d** $10x + 7y - 2x - y$

2.05

8 Evaluate each expression.

a $1.35 + 0.2 - 5.78$ **b** $35.3 - 12.91$

2.06

9 Evaluate each expression.

a 0.8×9 **b** 1.5×0.3 **c** $2.1 \div 5$ **d** $5.52 \div 0.3$

3.02

10 Write each statement as an algebraic expression.

a 5 more than k **b** 20 times k

c half of k **d** 3 lots of k increased by 10

e the difference between k and 12 where k is larger **f** k multiplied by itself

3.05

11 Simplify each expression.

a $5 \times 2p$ **b** $12m \div 2m$ **c** $-4x \times 3xy$

d $3a \times 2b \times 4$ **e** $\frac{14xy}{7xy}$ **f** $\frac{5a \times 4a}{10a}$

2.10

12 Use a factor tree to write each number as a product of its prime factors, in index notation.

a 180 **b** 875

2.10

13 Use factor trees to find the highest common factor of each pair of numbers.

a 28 and 72 **b** 36 and 90

2.11, 2.12

14 Use index notation to simplify each expression.

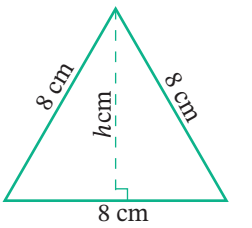
a $2^4 \times 2^3$ **b** $7^6 \times 7^5$ **c** $5^8 \div 5^3$ **d** $8^5 \div 8^4$

e $(3^4)^2$ **f** $(11^3)^5$ **g** 7^0 **h** $(8 - 5)^0$

1.08

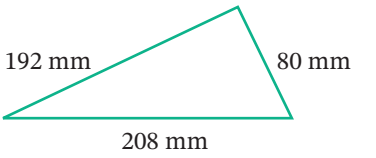
15 Find the height, h cm, of an equilateral triangle with sides 8 cm long. Write your answer:

a as a surd **b** correct to 2 decimal places.



1.06

16 Test whether this triangle is right-angled. If it is, copy the triangle and draw the right angle.



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17 Use factor trees to find the lowest common multiple of each pair of numbers. 2.10

a 12 and 18 **b** 18 and 27

18 Expand and simplify each expression. 3.07

a $4(x + 2)$ **b** $6(m - 4)$ **c** $-7(2k + 1)$

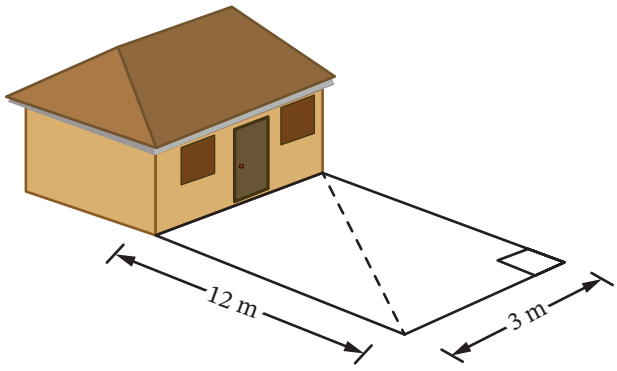
d $-(3p + q)$ **e** $2(m + 2) + 4m$ **f** $8 + 4(p - 2)$

19 Factorise each expression. 3.08

a $2m + 4$ **b** $d^2 - d$ **c** $6p + 12q$

d $-12x - 24$ **e** $2ab + 3a$ **f** $-2x^2 + 6x$

20 Rohan is laying a concrete slab 12 m by 3m in front of his shed. He needs to check that the corners of the slab are right angles. How long should the diagonal be? Answer correct to 2 decimal places. 1.07



Foundation Standard Complex